MATH 111 Calculus 1

Inverse functions

and their derivatives

Week 5 · Lecture 11 · §6.1

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. $3^3 = 27$. Without a calculator, find $\sqrt[3]{28}$ to two decimal places.

STAND UP

"$f(x) = x^2$ has an inverse function, and $f^{-1}(x) = \sqrt{x}$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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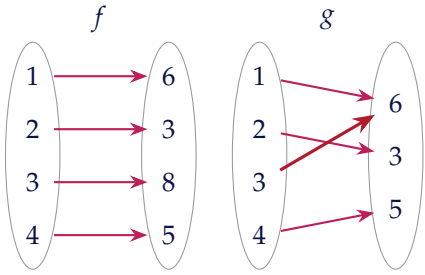
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Verdict

FALSE. x^2 takes the value 9 twice, $f(3) = f(-3) = 9$, so $f^{-1}(9)$ would have to be both 3 and -3. And the square root does not undo squaring for negative numbers: $\sqrt{(-3)^2} = 3 \neq -3$. $\sqrt{x}$ is the inverse of x^2 restricted to $[0, \infty)$ (frame 7).

One-to-one functions

To run a function *backwards* means: from each output, go back to the input it came from. When is that possible?

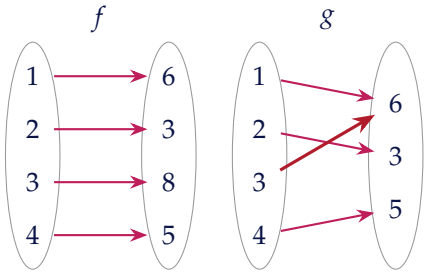


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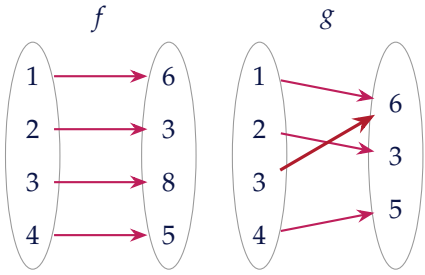
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Definition

A function f is one-to-one if it never takes the same value twice:

$$f(x_1) \neq f(x_2) \quad \text{whenever} \quad x_1 \neq x_2.$$

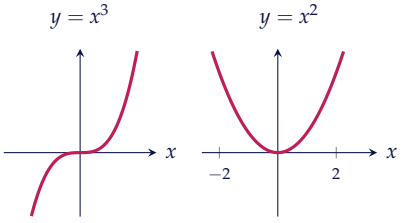


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The horizontal line test

Horizontal line test

f is one-to-one $\iff$ no horizontal line meets the graph of f more than once.

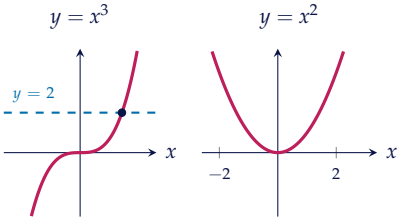


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x^3 is one-to-one: it is increasing — if $x_1 < x_2$, then $x_1^3 < x_2^3$. Why:
$x_2^3 - x_1^3 = (x_2 - x_1)(x_1^2 + x_1x_2 + x_2^2)$, and the second factor, $\left(x_1 + \frac{x_2}{2}\right)^2 + \frac{3}{4}x_2^2$, is positive.

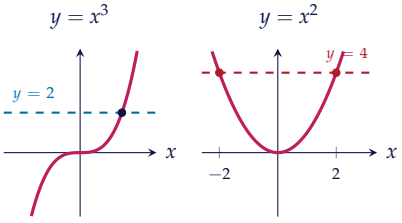


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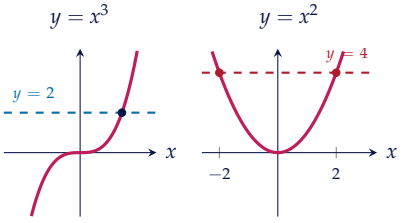
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An increasing function ($x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$) is one-to-one: different inputs, different outputs; so is a decreasing one. $x^3 + 2x$ is increasing: x^3 and $2x$ both are.

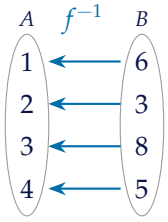


The inverse function

Definition

Let f be one-to-one, with domain A and range B. Its inverse function f^{-1} has domain B and range A — the roles swap — and for every y in B:

$$f^{-1}(y) = x \quad \iff \quad f(x) = y.$$



frame 3's f, backwards:
$f^{-1}(8) = 3$

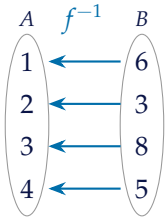
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Each undoes the other.



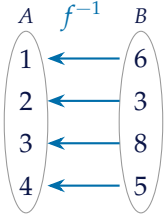
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Caution

$f^{-1}(x)$ does *not* mean $1/f(x)$: the -1 is not a power of the number $f(x)$. For the cube, $h(x) = x^3$: $h^{-1}(8) = 2$, but $1/h(8) = 1/512$. The reciprocal is written $\big[f(x)\big]^{-1}$.

Finding a formula for f^{-1}

Three steps

1. Write $y = f(x)$. **2.** Solve for x in terms of y. **3.** Swap x and y: the result is $y = f^{-1}(x)$.

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Step 2. $y(x-2) = x+1$, so $x(y-1) = 2y+1$, so $x = \dfrac{2y+1}{y-1}$, $y \neq 1$. If $f(x) = y$, then $y \neq 1$ ($y = 1$ would give $0 = 3$) and x is this one number: at most one x for each y, so f is one-to-one. Conversely, for each $y \neq 1$ this x is not 2, and the steps reverse: $f(x) = y$.

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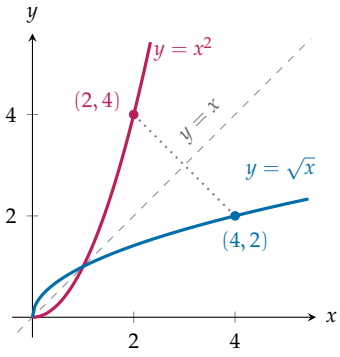
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Check with one number: $f(3) = \frac{4}{1} = 4$ and $f^{-1}(4) = \frac{9}{3} = 3$. Undoing, as promised.

The graph of f^{-1}

(a, b) is on the graph of $f \iff f(a) = b \iff$
$f^{-1}(b) = a \iff (b, a)$ is on the graph of f^{-1}.



$y = x^2$ $(x \geq 0)$ and $y = \sqrt{x}$: mirror images in $y = x$

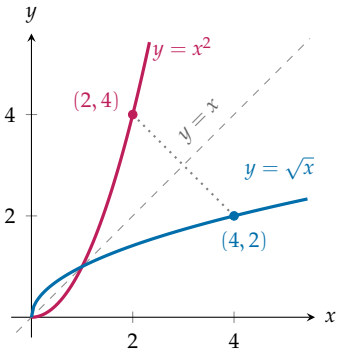
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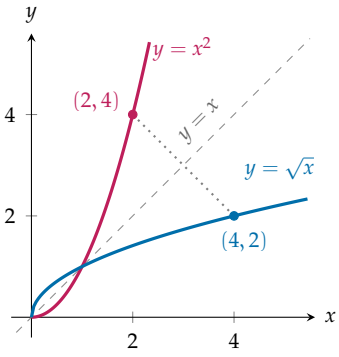
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Back to the warm-up. On $[0, \infty)$ the square x^2 is one-to-one, and its inverse is the square root $\sqrt{x}$: $\sqrt{x^2} = x$ for $x \geq 0$. The point $(2, 4)$ on one graph, $(4, 2)$ on the other.



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Question 31

Let $f(x) = 7 - 2x^3$. Which function is $f^{-1}(x)$?

A. $\sqrt[3]{\dfrac{x-7}{2}}$

B. $\dfrac{1}{7-2x^3}$

C. $\sqrt[3]{\dfrac{7-x}{2}}$

D. $\dfrac{\sqrt[3]{7-x}}{2}$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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60 sec

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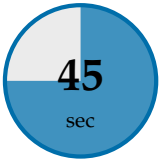
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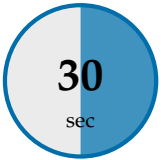
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Hands down. Thank you.

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Answer

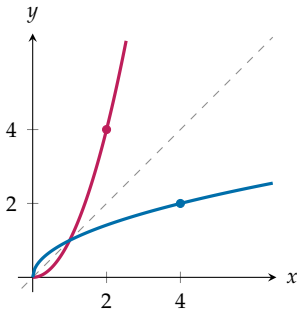
C. $y = 7 - 2x^3 \Rightarrow 2x^3 = 7 - y \Rightarrow x = \sqrt[3]{(7-y)/2}$; swap x and y. Check with one number: $f(1) = 5$, and only C sends 5 back to 1: $\sqrt[3]{(7-5)/2} = 1$. A moves the 7 with the wrong sign (-1 at 5). B is $1/f$ — the -1 read as a power. D divides by 2 after the root ($\sqrt[3]{2}/2 \approx 0.63$ at 5).

From the picture: slopes turn upside down

Theorem

Let f be one-to-one and continuous, with domain an interval I. Then $f(I)$ is an interval (open if I is open), and f^{-1} is continuous on it.

The picture: the reflection of an unbroken graph is unbroken. The proof uses the IVT (Lecture 5); we take this theorem without proof.



slope 4 at $(2, 4)$; slope $\frac{1}{4}$ at $(4, 2)$

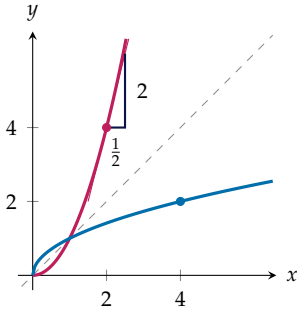
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Reflection in $y = x$ swaps x and y, so it swaps run and rise. At $(2, 4)$, $y = x^2$ has slope 4 (run $\frac{1}{2}$, rise 2).



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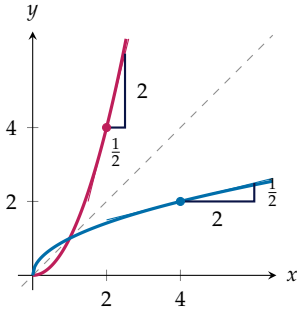
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The picture: the reflection of an unbroken graph is unbroken. The proof uses the IVT (Lecture 5); we take this theorem without proof.

Reflection in $y = x$ swaps x and y, so it swaps run and rise. At $(2, 4)$, $y = x^2$ has slope 4 (run $\frac{1}{2}$, rise 2).

Reflected, at $(4, 2)$: $y = \sqrt{x}$ has run 2, rise $\frac{1}{2}$ — slope $\frac{1}{4}$.



slope 4 at $(2, 4)$; slope $\frac{1}{4}$ at $(4, 2)$

From the picture: slopes turn upside down

Theorem

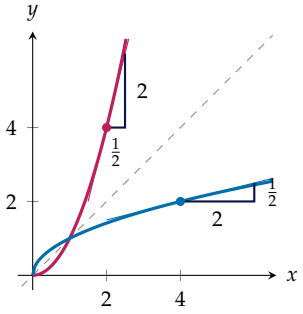
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In general, if $f(b) = a$ and $f'(b) \neq 0$:
$(f^{-1})'(a) = 1/f'(b)$. A picture, not a proof (frame 13).



slope 4 at $(2, 4)$; slope $\frac{1}{4}$ at $(4, 2)$

The derivative of an inverse function

Theorem

Let f be one-to-one and differentiable, with domain an open interval I; let b be in I, and let $a = f(b)$. If $f'(b) \neq 0$, then f^{-1} is differentiable at a, and

$$(f^{-1})'(a) = \frac{1}{f'(b)} = \frac{1}{f'\big(f^{-1}(a)\big)}\,.$$

A function defined on more than I is first restricted to I — as x^2 to $(0, \infty)$ on frame 11.

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How to read it. First find the point $b = f^{-1}(a)$ — the x at which f *equals* a. Then

$$(f^{-1})'(a) = \frac{1}{\underbrace{f'}_{\substack{\text{the derivative} \\ \text{of } f}} \quad \underbrace{\big(f^{-1}(a)\big)}_{\substack{\text{evaluated at } f^{-1}(a), \\ \text{not at } a}}}.$$

The derivative of an inverse function — the proof

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$$(f^{-1})'(a) = \lim_{x \to a} \frac{f^{-1}(x) - f^{-1}(a)}{x - a} = \lim_{y \to b} \frac{y - b}{f(y) - f(b)} = \lim_{y \to b} \frac{1}{\dfrac{f(y) - f(b)}{y - b}}$$

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$$= \frac{1}{\lim\limits_{y \to b} \dfrac{f(y) - f(b)}{y - b}} = \frac{1}{f'(b)} = \frac{1}{f'\big(f^{-1}(a)\big)} \qquad \square$$

Note that $x \neq a \Rightarrow f(y) \neq f(b)$ because f is one-to-one.

The chain rule gives the same formula — and shows where it fails

If we already *know* that f^{-1} is differentiable, the chain rule gives the formula: differentiate $f\big(f^{-1}(x)\big) = x$:

$$f'\big(f^{-1}(x)\big) \cdot (f^{-1})'(x) = 1. \qquad (*)$$

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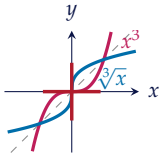
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Where it fails

If $f'(b) = 0$, then f^{-1} is *not* differentiable at $a = f(b)$.
Proof. Otherwise $(*)$, at $x = a$, would read $0 \cdot (f^{-1})'(a) = 1$. $\square$ Example: $(x^3)' = 0$ at 0, and $\sqrt[3]{x}$ has a vertical tangent there.



red: the two tangents at 0

Two examples

(a) The cube root. $f(x) = x^3$ is one-to-one and differentiable on $\mathbb{R}$, $f'(b) = 3b^2 \neq 0$ for $b \neq 0$, and $f^{-1}(a) = \sqrt[3]{a}$. So for $a \neq 0$:

$$(f^{-1})'(a) = \frac{1}{f'\left(\sqrt[3]{a}\right)} = \frac{1}{3\left(\sqrt[3]{a}\right)^2};$$

for $a > 0$ this is $\frac{1}{3}\, a^{-2/3}$: Lecture 9's power rule with $n = \frac{1}{3}$ — stated there, proved here. (At 0: $f'(0) = 0$, and the vertical tangent of frame 14.)

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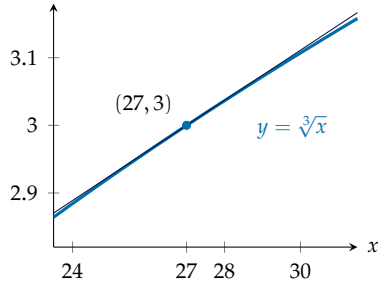
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We never solved $y = x^3 + 2x$ for x — and we never needed to.

Back to the title slide: $\sqrt[3]{28}$

$f(x) = x^3$: $f(3) = 27$ and $f'(3) = 27$. So $(f^{-1})'(27) = \dfrac{1}{27}$: near 27, the cube root rises by about $\frac{1}{27}$ for every unit.

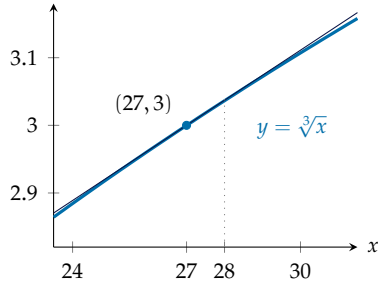


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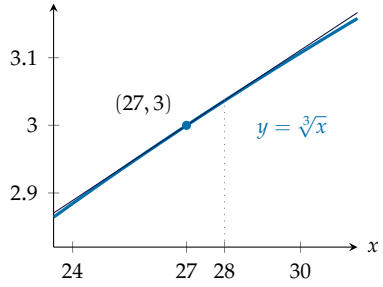
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Is the second decimal right? Still without a calculator:
$3.035^3 = 27.956\ldots < 28 < 28.233\ldots = 3.045^3$, so $\sqrt[3]{28} = 3.04$ to two decimal places. (The calculator: $3.03658\ldots$)



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Question 32

f is one-to-one and differentiable on $\mathbb{R}$, and $f(1) = 4$, $f'(1) = 3$, $f(4) = 10$, $f'(4) = 5$. What is $(f^{-1})'(4)$?

A. $\frac{1}{5}$

B. 3

C. $-\frac{1}{20}$

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In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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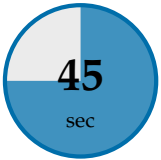
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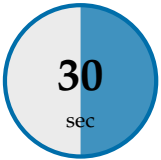
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Hands down. Thank you.



Question 32

EXPLAIN

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Answer

D. $f(1) = 4$, so $f^{-1}(4) = 1$, and $(f^{-1})'(4) = \dfrac{1}{f'(f^{-1}(4))} = \dfrac{1}{f'(1)} = \dfrac{1}{3}$. A is $1/f'(4)$: f' fed 4 instead of $f^{-1}(4) = 1$. B is $f'(1)$: the reciprocal forgotten. C is the derivative of $1/f$ at 4, $-f'(4)/f(4)^2 = -5/100$: the -1 read as a power. $f(4) = 10$ is not needed for the answer (only C uses it).

A little history · 1813–1820

1813. In a paper in the *Philosophical Transactions* of the Royal Society, **John Herschel** (born 1792) writes $\cos^{-1}$ for the inverse of the cosine — and adds a footnote: the notation "must not be understood to signify" one over the cosine.

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By 1820 he had learned that a German analyst, **Bürmann**, had explained the notation before him — though, Herschel adds, Bürmann seems not to have applied it to $\tan^{-1}$ and the other inverse functions. It stayed — and so did the danger of frame 5: $\sin^2 x$ is a square, but $\sin^{-1} x$ is an inverse (Lecture 14).

Today in two lines

1. A one-to-one function can be run backwards: $f^{-1}(y) = x \iff f(x) = y$. Domain and range swap; the graph is reflected in $y = x$; and $f^{-1}(x)$ is not $1/f(x)$.

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This week:

Homework: §6.1 — read and work out the examples. Next: exponential functions and their derivatives (§6.2).

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"To find* $(f^{-1})'(4)$ *I need* f' *at* $f^{-1}(4)$*, not at* 4*, because . . . " (one sentence, or a picture).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.